

Chapter 100 — Lower Urinary Tract Calculi

HISTORY

- **Hippocrates' oath** reserved "cutting for stone" to specialists — the earliest statement of urological subspecialisation.
- **Ancient disease** in Greek, Egyptian and Indus Valley civilisations; the **earliest known lower tract stone** is from the Egyptian grave of a 16-year-old male, c. 4800 BCE.
- **Three classical routes** to the bladder stone: perineal, suprapubic, transurethral.
- **"Lithotomy"** was coined by Ammonius, c. 276 BCE; **Celsus**, some 300 years later, recorded it and performed the first perineal urethrotomy in the lithotomy position — the **methodus Celsiana**.
- **Apparatus minor versus major**: minor used no instruments, the stone cut down upon through a perineal incision beside the anus; major (**Marianus Sanctus**, Italy) used many.
- **Archaeological bladder stone** in a middle-aged man from a British excavation was attributed to bladder outlet obstruction (BOO).
- **Jean Civiale** developed **lithotrity** (transurethral stone crushing); his student **Sir Henry Thompson** operated on King Leopold I.
- **Paradigm shift**: vesical calculi have steadily declined in the West over 50 years but persist in developing nations, **particularly in children**.

Concept — *Bladder stone has shifted from a paediatric nutritional disease to an adult disease of outlet obstruction and foreign material wherever diet has changed; the endemic paediatric form survives where it has not.*

ETIOPATHOGENESIS OF BLADDER CALCULI

Epidemiology and classification

- **Commonest lower tract stone**: about 5% of all urinary stone disease and 1.5% of urological admissions in the West.
- **Non-endemic areas**: adults, secondary to another disease. **Endemic areas**: children, without associated anomaly, driven by **diet and socioeconomic conditions**.
- **Classification**: migrant, primary idiopathic (no predisposing cause) and secondary (predisposing cause present).

PRIMARY VERSUS SECONDARY BLADDER CALCULI

	Primary (endemic)	Secondary
Setting	Undernourished populations; North Africa, Middle East	Western world
Age and sex	Children : form within first 5 years; commonest under 10, peak 2–4 years; male 9:1 to 33:1	Men over 60 with lower urinary tract obstruction
Cause	Diet : low animal protein, low phosphate, high oxalate; vitamin deficiency	Bladder pathology : BOO, foreign body, catheter, infection
Associated anomaly	None	Always a predisposing cause
Composition	Ammonium acid urate , calcium oxalate, uric acid, calcium phosphate	Struvite predominant

	Primary (endemic)	Secondary
Number / recurrence	Solitary; rarely recurs	Recurr unless cause removed

- **What it tests:** that "bladder stone" is two diseases — one of the urine, one of the bladder. **Trap:** looking for an outlet problem in the endemic child.

Primary stone — mechanism

- **Decreased urine output + altered urine pH + metabolic abnormality** on a background of animal-protein deficiency.
- **Cereal-based, low-phosphate diet** → **hypophosphaturia** → precipitation of calcium oxalate and ammonium acid urate; an **oxalate-rich diet** adds hyperoxaluria.

Secondary stone — causes

- **Origin:** de novo, or migrant from the upper tract and retained. The frequent **absence of a calcium oxalate nucleus** argues against upper tract origin for most.
- **Significant postvoid residual (PVR):** BOO from stricture or benign prostatic hyperplasia (BPH), neurogenic bladder, urinary diversion, bladder neck contracture.
- **Foreign bodies:** migrated intrauterine devices, pessaries and diaphragms, eroded clips. Uncommon: eroded cerclage wire, unrecognised anorectal impalement, eroded silk sutures from the dorsal vein complex, migrated brachytherapy seeds, **hair introduced during clean intermittent catheterisation (CIC)**.
- **Nonabsorbable suture migrates** to the bladder; erosion with stone after Stamey procedure, open colposuspension and prosthetic retropubic sling.
- **Renal transplantation:** stones form on **both absorbable and nonabsorbable** sutures at ureteroneocystostomy.

Catheters and nucleation

- **Chronic indwelling catheters:** stone incidence 0.7–2.2%; **retained fragments of burst Foley balloons** act as a nidus.
- **Heterogeneous nucleation** around a foreign body, from supersaturation due to infection or obstruction, is the mechanism; **homogeneous nucleation is rarely responsible**.

Neurogenic bladder

- **Spinal cord injury (SCI)** carries the highest risk; risk factors: complete lesion, **Klebsiella urinary tract infection (UTI)**, White race.
- **Hypothesis:** chronic mucosal injury and inflammation from catheter or stone → **glycosaminoglycan (GAG) layer deficiency** → stone.
- **Large series:** struvite predominant; uric acid stones **only in males**.

Drugs

- **Triamterene** (potassium-sparing diuretic acting on distal tubular sodium reabsorption) forms stones by **precipitation of drug and metabolites**, reported with BOO and high residuals. **Indinavir** (protease inhibitor) is also lithogenic.

Concept — Primary bladder stone is a disease of the urine — diet-driven supersaturation in a normal bladder; secondary stone is a disease of the bladder — stasis, a foreign nidus and infection. This is why the first rarely recurs and the second recurs unless its cause is removed.

PRESENTATION OF BLADDER STONES

- **Rarely asymptomatic. Commonest presentation: terminal haematuria.**

PRESENTATION — CHILDREN VERSUS ADULTS

Children (primary stone)	Adults
Rarely acute ; prodrome of cloudy, sandy urine	Terminal haematuria with lower urinary tract symptoms (LUTS): intermittency, frequency, urgency, poor flow, urge incontinence, abdominal pain
Abdominal discomfort, dysuria, frequency, haematuria; acute retention rare	May present with acute urinary retention

MANAGEMENT OF BLADDER STONES

- **Options:** medical management, extracorporeal shock wave lithotripsy (ESWL), transurethral lithotripsy, percutaneous cystolithotomy and cystolitholapaxy, open surgery.
- **Deciding factors:** stone size and composition, patient age and build, **coexistent stones elsewhere**, **concomitant BOO**, available expertise and equipment.

Medical Management

- **Chemodissolution alone** is slow and incompletely effective — now an **adjunct in select cases**, most effective for **encrustation on long-term catheters** (treatment and prophylaxis).

Extracorporeal Shock Wave Lithotripsy

- **Indicated where endoscopy threatens a device** — artificial urinary sphincter, penile prosthesis — and for **neobladder stones** and **medically high-risk** patients.
- **Poor choice in continent reservoirs:** fragments often need endoscopic removal.
- **Protocol:** prone (avoids coccygeal and rectal artefact) → three-way Foley → fill to **150 mL** to localise → treat, with intermittent irrigation aiding localisation → drain.
- **Outcome determinants:** PVR, composition, size.

Endourologic Approach to Bladder Stone

- **Miniaturised equipment** with efficient energy sources means the **majority of bladder stones are now treated endourologically**.
- **Terms:** **cystolithotomy** — intact removal; **cystolithotripsy** — fragmentation with an energy source; **cystolitholapaxy** — mechanical breakage.

TRANSURETHRAL VERSUS PERCUTANEOUS CYSTOLITHOTRIPSY

	Transurethral	Percutaneous
Status	Most popular (ease, robust energy); laser the modality of choice — single stage, minimal complications	Reserved for specific indications
Indication	Adequate urethra	Difficult urethra: previous bladder neck reconstruction or closure
Energy / access	Holmium laser, electrohydraulic lithotripsy (EHL), pneumatic lithoclast	Amplatz sheath or Hassan cannula; ultrasonic + pneumatic ; entrapment sac for fragments
Key risk	Urethral stricture from repeated instrument passage	Saves the urethra ; urine leak, haematuria uncommon; high single-sitting success

- **Stricture prevention** transurethrally: transurethral Amplatz sheath, adequate lubrication, preoperative meatotomy, postoperative suprapubic catheter.
- **Mechanical lithotrite is falling from favour** (mucosal injury, perforation), although recent mechanical cystolitholapaxy reports very high stone-free rates.

Figure 100.1 — Large bladder stone and cystolitholapaxy equipment

- (A) Large bladder stone. (B) **Stone-crushing forceps** — 25 Fr shaft for telescope, serrated jaws. (C) **Optical stone-crushing forceps** — stone broken **smaller than the 25 Fr sheath**. (D) **Mauermayer's stone punch**. (Sabnis, Patwardhan, Ganpule, *Urology Instrumentation: Step by Step Guide*, 2016.)

Open Surgery for Bladder Stones

- **Prolonged catheterisation and hospital stay**; catheterless, drainless two-layer cystolithotomy is feasible in children.
- **Indicated where transurethral or percutaneous access is contraindicated**: small-capacity bladder, urethral stricture.

Concept — Modality is dictated less by the stone than by the route to it: an adequate urethra permits transurethral laser; a compromised urethra or an implanted device diverts to percutaneous access or ESWL; a small bladder or no safe route leaves open surgery.

LOWER TRACT CALCULI IN SPECIAL SITUATIONS

Bladder Outlet Obstruction With Bladder Lithiasis

- **BOO causes incomplete emptying and non-evacuation of fragments**, present in 45–79% of patients with vesical calculi.
- **Bladder stone secondary to BOO is an absolute indication for BPH surgery**.
- **Infection also contributes** — over 30% of bladder stones are infected, usually by **urea-splitting organisms** — so **not every secondary stone requires outlet surgery**. Evidence conflicts: a urodynamic study before and after stone treatment found BOO in **only about half**; another series found BOO the cause in the **great majority**.

Figure 100.2 — Bladder calculus, one patient

- Plain radiograph; ultrasonography supine and lateral — the stone **shifts with position**; cystoscopic view.

Concept — Obstruction and infection are two independent routes to the secondary bladder stone; outlet surgery is mandatory when the stone is due to BOO, so the examinable point is proving that BOO, not assuming it from the stone.

Bladder Calculi in Urinary Diversions and Augmented Bladder

- **Upper and lower tract stones** occur after augmentation or diversion; incidence varies with the operation. **Three aetiological groups**: infection, metabolic, anatomical.
- **Metabolic**: ileum or colon → **hyperchloraemic metabolic acidosis** → hypercalciuria (reduced renal tubular calcium reabsorption); **loss of ileum** → **enteric hyperoxaluria** and diarrhoea.
- **Infection**: stones **mainly struvite**; urease producers — *Pseudomonas*, *Proteus*, *Klebsiella* — alkalinise urine and promote crystallisation. **Mucus** aids biofilm and **impairs antibiotic penetration**.
- **Stomach reduces stone incidence**, but haematuria-dysuria syndrome and long-term proton pump inhibitor (PPI) need discourage its use.
- **Stasis**: stomal stenosis (conduit) or poor emptying (continent diversion). **Absorbable sutures** significantly reduced stones.
- **Know diversion type and anatomy** preoperatively: it determines modality, guides percutaneous puncture and avoids **bowel and vascular pedicle injury**.

APPROACH BY RECONSTRUCTION

Reconstruction	Approach	Reason / caveat
Conduit, stomal stenosis	Clear stone + revise stoma	Recurr while obstruction persists

Reconstruction	Approach	Reason / caveat
Continent cutaneous pouch	Percutaneous	Avoid injury to continence mechanism
Neobladder, small burden	Transurethral	Minimal sphincter injury
Neobladder, large burden	Percutaneous or open	Percutaneous via Amplatz sheath or laparoscopic trocars

- **Postoperative drainage:** suprapubic or stomal catheter for stoma or pouch; urethral catheter for neobladder.
- **Prevention:** complete clearance, treat infection, correct anatomy, **prevent mucus accumulation**.

Concept — The diversion stone forms where bowel mucosa (acidosis, mucus, biofilm) meets stasis; treatment must spare the continence mechanism and vascular pedicle, and prevention targets mucus, infection and suture.

Bladder Calculi in Patients With Spinal Cord Injury

- **Neurogenic bladder** from SCI or myelomeningocele raises risk; determinants: level and severity of lesion, previous stones, persistent infection, **drainage method**.
- **Risk peaks 3 months after injury**; 15–30% form a stone within 10 years; **risk quadruples after the first stone**. Alkaline urine with **hypocitraturia** contributes.

THE TWO PHASES AFTER SPINAL CORD INJURY

Acute phase (immobilised)	Recovery phase
Immobilisation hypercalciuria → calcium stones anywhere, kidney included	Persistent infection from inadequate emptying or drainage method

- **CIC has lower stone incidence than an indwelling catheter**; suprapubic and urethral indwelling catheters carry the same incidence.
- **Avoid indwelling Foley** (encrustation, nephrolithiasis); if unavoidable, **weekly changes** dramatically reduce stones.
- **Prevention:** long-term antibiotics, **ascorbic acid to acidify urine**, fluids, antibacterial bladder lavage, silicone catheters.
- **Treatment:** up to 4 cm → **transurethral laser**; larger → percutaneous or open. **ESWL is ideal for anaesthetic risk or autonomic dysreflexia**.

Concept — SCI stone risk follows the two phases of injury — metabolic early, infective late — and the drainage method is the modifiable lever: CIC beats indwelling catheterisation, but suprapubic offers no advantage over urethral.

Bladder Calculi After Renal Transplantation

- **Post-transplant urolithiasis** 1–1.8%; renal stones commoner, but bladder calculi quite common.
- **Causes:** persistent **Proteus** infection, outflow obstruction, **retained suture**; calcineurin inhibitors and steroids contribute. Absorbable suture has reduced incidence.
- **First investigation:** **ultrasonography**; computed tomography (CT) to exclude renal or ureteric calculi.
- **Treatment as routine** — cystolitholapaxy, laser, ESWL; **removal of all exposed intraluminal suture is paramount**.

Concept — The transplant bladder stone is a suture-and-infection stone; the modality is routine, and clearing exposed suture is the step that prevents recurrence.

Treatment of Vesical Calculus in Children

- **Options as in adults:** ESWL, open cystolithotomy (OC), percutaneous cystolithotripsy (PCCL), transurethral cystolithotripsy (CL); energy: pneumatic, holmium laser, thulium fibre laser.

PAEDIATRIC SERIES — APPROACH AND VERDICT

Approach and energy	Population	Verdict
PCCL; fluoroscopic tract dilatation; ultrasonic	Endemic stones	Safe, credible option
PCCL	Infants	Successful
Transurethral 8 Fr ureteroscope, Ho:YAG (0.6–1.8 J, 5–12 Hz, 550 µm end-firing fibre)	Children 2–12 years	Safe and efficient
PCCL; access into distended bladder without imaging; pneumatic	Endemic stones >1 cm; largest single-centre series	No significant complications

- **Transurethral limitation:** the **small-calibre male urethra** limits large stones and risks **iatrogenic stricture**.

Concept — In children the urethra, not the stone, is the constraint: percutaneous access spares it and is the route for large stones, while transurethral laser is kept to what a small urethra can safely take.

URETHRAL CALCULI

- **Rare:** 0.3–1% of all stones; an **obstructing urethral calculus is very rare**. A rarity in the West, not uncommon in endemic Asia and the Middle East; declining overall.
- **Males; bimodal age:** early childhood and fourth to fifth decades. **Short female urethra** and **higher peak flow** in the young are protective.

Pathogenesis and Composition of Urethral Calculi

- **Commonest site:** **posterior (prostatic) urethra**; less commonly pendulous.
- **Vast majority migratory:** stricture, meatal stenosis or BPH arrests a stone that would otherwise pass.
- **Upper tract, not bladder, is often the source:** composition mostly calcium oxalate, as upper tract stones; concomitant stones commoner in the upper tract than the bladder; in endemic areas bladder calculi fell with dietary change while urethral calculi did not.

Primary Urethral Calculi

- **Form at their final site**, secondary to obstruction or **stasis in a diverticulum**; **urethral stricture is the leading cause**. Organisms: *Escherichia coli*, *Proteus*, enterococci.
- **Hair-bearing grafts** (stricture or hypospadias repair): hair is a nidus; stones adhere to hair ball and mucosa. **Epilation reduces but does not eliminate risk** — follicles persist.
- **Foreign bodies:** **brachytherapy seeds** migrating into the prostatic urethra; **cryoablation** necrosis with persistent inflammation.

MIGRATORY VERSUS PRIMARY URETHRAL CALCULI

	Migratory (secondary)	Primary (autochthonous)
Frequency	Vast majority	Minority
Source	Upper tract or bladder	In situ: stricture, diverticulum, hair, foreign body
Composition	Calcium oxalate (85–90%)	Struvite ; also calcium phosphate, uric acid
Morphology	Solitary, larger ; steinstrasse reported	Small, round, no core or nucleus
History	Stone disease, instrumentation, flank pain 1–2 weeks before	Insidious; voiding pain, obstructive LUTS, chronic pelvic pain, recurrent UTI; women: frequency, occasional incontinence
Timing	Present immediately	Present late , often years
Urine	Haematuria	Haematuria + infection

Concept — The urethral stone is usually an upper tract stone arrested by a narrowed urethra; the rarer primary stone is a stasis-and-infection stone of stricture, diverticulum or hair-bearing graft.

Clinical Presentation and Evaluation

- **Commonest presentation:** acute painful urinary retention from sudden impaction. If still voiding: weak, interrupted or splayed stream, gross haematuria, dysuria.
- **Pain localises the stone:** posterior → perineal or rectal; pendulous → penile tip.
- **Posterior stones are commoner** (up to 88%) than anterior.
- **Examination:** posterior — often normal bar a mildly tender prostate; bulbar or penile — **palpable hard mass**; female — hard mass in the **anterior vaginal wall**; large diverticula palpable with multiple perineal stones.
- **Delay** → larger stones and **urethrocutaneous or urethrorectal fistula**, especially with reduced sensation (infants, SCI).
- **About 98% radio-opaque** (older literature 40%) → plain radiograph.
- **Retrograde urethrogram (RUG):** filling defect; historical mainstay, now **reserved for inconclusive ultrasonography or suspected stricture**.
- **Transrectal ultrasonography (TRUS):** hyperechoic with **posterior acoustic shadowing**; a 10 MHz linear probe along the dorsal penis screens the whole urethra.
- **Image the upper tract** (ultrasonography or CT) for coexistent stones.

Treatment

- **Determinants:** location and distance from either meatus, stone characteristics, **whether it can be pushed back**, associated urethral abnormality.

URETHRAL CALCULI — TREATMENT BY SITE

Site	First choice	Escalation
Posterior	Push back → intracorporeal laser (least traumatic, 85–90% success)	Not pushable → in situ laser or pneumatic → open cystolithotomy last
Anterior (pendulous, bulbar)	No pushback ; in situ holmium laser; milking if smooth	Urethrotomy for large, proximal stones
Fossa navicularis / distal	Forceps under anaesthesia; lidocaine jelly for small stones	Ventral meatotomy
Diverticular	Incision and extraction	Diverticulectomy + repair, same or staged
With stricture / fistula	Simultaneous or staged repair	

- **ESWL after pushback:** historical; low success with residual stones. **Micropercutaneous** instruments: less bulky and traumatic, own learning curve, equivalence unproven.
- **Anterior pushback is hardly ever possible and should not be attempted.** Milk only smooth stones — never large or rough ones (urethral injury).
- **In situ EHL or Swiss LithoClast** succeed up to 80%; urethral damage concerns are offset by **holmium laser**. **Ventral meatotomy** allows controlled removal with less trauma.

Figure 100.3 — Urethral calculi

- Calculus **impacted at the bulbomembranous junction**; laser fragmentation.

Concept — The governing variable is whether the stone can go back: posterior stones return to the bladder and are treated there; anterior stones cannot, and are extracted or fragmented where they sit, with invasiveness escalating with size, roughness and distance from the meatus.

Unusual Lower Tract Urolithiasis

Preputial Calculi

- **Adults** with **severe phimosis**, **poor hygiene**, **low socioeconomic status**.
- **Mechanism**: smegma inspissation, stasis with salt precipitation, or both — a **vicious cycle** of scarring and worsening phimosis. Also foreign bodies, suture, **extruded bladder calculi**.
- **Presentation**: LUTS, **difficulty voiding commonest**; dysuria, haematuria, foul urine, palpable stones; occasionally retention; **tight phimosis**. Lymphadenopathy → plain radiography ± CT.
- **Treatment**: **circumcision or dorsal slit**; **excised skin to histopathology**. Stones multiple, smooth, rounded, struvite.

Concept — *Phimosis turns the preputial sac into a stagnant, infected reservoir; removing the prepuce removes the reservoir, and the excised skin always goes for histology.*

Prostatic Calculi

- **Association with LUTS** recognised in the early 20th century; **TRUS** detects them increasingly; autopsy shows calcification in **almost all asymptomatic adult men**.
- **Prostatic urethral calculi** are migratory — **manage as urethral calculi**. **True prostatic calculi** lie within the gland.

TRUE PROSTATIC CALCULI — ENDOGENOUS VERSUS EXOGENOUS

Primary / endogenous	Secondary / exogenous
Within the acini ; more numerous, smaller	Reflux of urine ; fewer, larger

- **Composition**: **calcium phosphate (majority)**, then calcium carbonate phosphate, calcium oxalate, mixed.
- **Commoner in histologically proven BPH**; **no correlation with adenocarcinoma**.
- **A prostatitis symptom** → **3.2 times more likely** to harbour a significant calculus; **size, not number**, correlates with symptoms.
- **Chronic pelvic pain syndrome (CPPS)**: calculi common, with more inflammation, more gram-negative culture positivity (*E. coli*, *Klebsiella*) and more white cells in expressed prostatic secretions; other work finds **no conclusive link**.
- **TRUS is the main modality**. **Harada classification**: **type A** — discrete, small, evenly distributed echoes; **type B** — coarse, larger, focal echoes.
- **Plain radiograph detects 14%**; role of CT and magnetic resonance imaging (MRI) unclear.
- **Calculi met at transurethral resection of prostate (TURP)** → **remove with the loop**: greater symptom relief.

Concept — *Locate first: a prostatic urethral stone is a urethral stone and is managed as one; a true prostatic calculus is a marker of glandular inflammation or reflux, not of cancer, and is removed when met at TURP.*

Seminal Vesicle Calculi

- **Seminal vesicles (SVs)**: paired glands posterior to the bladder; **calculi uncommon**.
- **Aetiologies**: haemospermia, gonorrhoea, prostatitis, vesiculitis, tuberculosis, UTI. **Symptoms**: none, or **haemospermia**, perineal or testicular pain, painful ejaculation, low volume.
- **Composition**: calcium phosphate, struvite, protein matrix.
- **Diagnosis**: vasoseminal vesiculography, MRI (**T2 hypointense, no diffusion restriction**), CT, **TRUS** — **most accurate**.

SEMINAL VESICLE CALCULI — SURGICAL OPTIONS

Option	Role
Laparoscopic or open seminal vesiculectomy; transurethral resection of ejaculatory ducts (TURED)	Standard therapy
Transurethral seminal vesiculoscopy (TRU-SVS)	6 Fr vesiculoscope , retrograde via verumontanum orifice → forceps or holmium laser; SV and ejaculatory duct stones; fewer complications than TURED
Transperitoneal laparoscopy	Large SV stone or failed ejaculatory duct cannulation

KEY POINTS

- Primary bladder calculi: children under 10, peak 2–4 years, far commoner in males.
- Bladder stones: commonest presentation haematuria with LUTS; children — abdominal discomfort, dysuria, frequency, haematuria.
- Options: medical, ESWL, transurethral, percutaneous cystolithotomy or cystolitholapaxy, open. Percutaneous: safe, effective, spares the urethra.
- Urethral calculi: 0.3–1% of stones; usually present with acute painful retention.
- Prostatitis symptom: 3.2 times more likely to harbour a prostatic calculus; plain radiograph shows prostatic calculi in 14%.
- Preputial calculi: severe phimosis, poor hygiene, low socioeconomic status.
- Seminal vesicle calculi: uncommon; consider in unusual LUTS.